

PASSO ROLLE, Italy

WMO: 160210

Lat: 46.297N

Lon: 11.787E

Elev: 6581

StdP: 11.52

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	6.1	9.1	-21.6	2.1	22.2	-16.0	2.9	21.1	27.7	24.4	23.2	24.0	5.2	330	0.720

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	66.6	54.9	64.4	54.4	62.3	53.4	58.5	62.9	56.7	61.3	55.2	59.7	8.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.9	88.4	60.6	55.0	82.4	59.2	53.3	77.4	57.7	28.6	63.8	27.2	61.4	26.1	59.7	67.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.9	18.0	15.4	DB	3.5	70.8	4.7	2.1	0.1	72.3	-2.6	73.5	-5.2	74.7	-8.6	76.2	(4)
(5)			WB	1.7	62.2	4.3	2.2	-1.4	63.8	-3.9	65.1	-6.3	66.4	-9.4	68.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.3	24.2	24.4	28.6	34.5	42.0	50.2	54.3	53.4	46.8	40.8	32.8	26.6	(6)	
	DBStd	12.48	6.55	7.87	7.20	5.70	6.06	6.34	4.78	5.05	5.48	6.30	7.17	7.29	(7)	
	HDD50	4672	799	717	664	464	258	76	17	25	123	289	515	725	(8)	
	HDD65	9747	1264	1137	1129	914	714	445	331	360	546	752	965	1190	(9)	
	CDD50	401	0	0	0	0	9	82	151	130	27	2	0	0	(10)	
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	6.1	6.2	5.5	6.2	5.9	6.5	6.5	6.6	6.6	5.9	5.5	5.8	6.1	(14)	
(15) Precipitation	PrecAvg	42.9	2.0	2.1	2.6	3.6	4.5	4.5	4.2	4.1	3.7	5.2	3.9	2.0	(15)	
	PrecMax	59.5	6.9	18.7	5.8	7.9	9.0	8.0	7.7	7.4	10.8	14.7	12.9	5.6	(16)	
	PrecMin	26.7	0.0	0.0	0.7	1.9	2.1	2.0	0.8	2.0	0.2	0.1	0.0	0.5	(17)	
	PrecStd	8.0	1.5	3.6	1.2	1.4	2.0	1.6	1.4	1.3	2.4	3.4	3.4	1.5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.3	58.6	46.8	51.6	62.2	68.4	69.9	69.4	62.8	56.8	51.7	44.7	(19)	
		MCWB	30.5	24.9	36.0	41.9	46.5	53.8	56.6	54.1	52.8	42.4	39.1	30.5	(20)	
	2%	DB	38.2	41.3	43.5	47.8	57.3	65.6	67.1	66.1	59.2	53.5	46.9	40.9	(21)	
		MCWB	29.8	30.0	34.1	39.5	47.5	53.7	56.0	54.7	50.8	42.1	37.4	29.7	(22)	
	5%	DB	35.4	37.4	40.8	44.8	53.9	62.6	64.5	63.0	56.9	50.4	44.1	38.0	(23)	
		MCWB	28.1	29.6	32.9	37.7	45.7	52.6	55.0	54.4	49.9	42.3	35.8	28.8	(24)	
	10%	DB	32.4	33.9	37.6	42.4	50.7	59.4	62.3	60.9	53.9	48.4	41.1	35.5	(25)	
		MCWB	26.8	27.0	31.8	36.6	44.0	51.5	53.8	53.4	48.3	41.6	34.1	28.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.1	33.9	38.3	44.3	51.9	59.4	61.7	60.3	56.3	49.2	42.8	34.9	(27)	
		MCDB	39.0	38.6	42.9	48.8	56.9	63.7	65.8	63.4	59.7	51.7	47.9	38.2	(28)	
	2%	WB	31.3	31.6	36.0	41.4	49.1	56.9	58.9	57.9	53.4	46.9	39.8	33.0	(29)	
		MCDB	35.1	36.1	40.4	45.6	54.3	61.9	63.6	62.2	56.7	49.6	44.7	36.1	(30)	
	5%	WB	29.7	29.4	34.2	39.1	47.3	54.7	57.1	56.1	51.3	45.1	37.7	31.3	(31)	
		MCDB	33.0	33.7	38.5	43.1	52.5	59.4	61.6	60.8	54.6	48.2	41.1	34.4	(32)	
	10%	WB	28.2	27.8	32.4	37.2	45.3	52.8	55.4	54.6	49.6	43.6	36.2	29.9	(33)	
		MCDB	31.2	31.7	36.4	40.4	49.9	57.7	59.9	59.3	52.9	46.7	38.9	33.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.4	10.7	10.6	10.7	11.7	12.5	12.8	11.9	10.6	9.4	8.0	8.0	(35)	
		MCDBR	10.4	16.5	12.1	12.5	14.5	14.7	14.2	13.2	12.0	11.1	10.0	9.6	(36)	
	5% WB	MCWBR	7.6	7.4	8.6	8.1	9.7	10.4	10.5	9.4	8.9	8.0	6.7	6.3	(37)	
		MCWBR	8.3	9.7	11.2	11.5	13.4	13.8	13.3	12.4	11.1	9.3	8.1	7.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.246	0.271	0.317	0.366	0.374	0.388	0.376	0.370	0.347	0.325	0.281	0.248	(40)	
		taud	2.429	2.379	2.322	2.241	2.263	2.259	2.294	2.328	2.375	2.417	2.496	2.490	(41)	
	Ebn at Noon		279	290	289	282	283	278	280	278	275	265	261	267	(42)	
		Edh at Noon	21	28	34	41	42	42	40	37	32	27	20	18	(43)	
(44) All-Sky Solar Radiation	RadAvg		495	782	1174	1421	1612	1836	1866	1618	1233	808	486	415	(44)	
	RadStd		58	98	146	187	173	129	126	140	118	104	80	52	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (66 neighbors)		+0.72	N/A	N/A	+1.13	+0.70	+0.61	N/A	-212	+139	+49

Nomenclature: See separate page